



MASENO UNIVERSITY
UNIVERSITY EXAMINATIONS 2017/2018

**THIRD YEAR FIRST SEMESTER EXAMINATIONS FOR THE
DEGREE OF BACHELOR OF SCIENCE AND BACHELOR OF
EDUCATION SCIENCE WITH INFORMATION TECHNOLOGY**

MAIN CAMPUS

SZL 301: DEVELOPMENTAL BIOLOGY

Date: 12th February, 2018

Time: 12.00 - 3.00 pm

INSTRUCTIONS:

- Answer ALL questions in SECTION A and any THREE questions in SECTION B.



SZL 301: DEVELOPMENTAL BIOLOGY

TIME: 2 Hours

Section A (40 Marks): Answer all questions. Each question carries 5 marks.

Q1: Justify the sentence "ontogeny recapitulates phylogeny".

Q2: Define the following terms

- a) Epigenesis
- b) Fertilization
- c) Neurulation
- d) Vitellogenesis
- e) Humunculus

Q3: Define the role of Hox genes in embryology.

Q4: State the role of the quantity and distribution of egg yolk in cleavage and gastrulation.

Q5: Explain patterning mechanisms of limb development in vertebrates.

Q6: Identify the THREE germ layers and list FOUR tissues or structures that develop from each.

Q7: Illustrate the major embryological processes following fusion of gametes.

Q8: Explain the five key questions in Developmental Biology.

Section B (30 Marks): Answer any TWO questions. Each question carries 15 marks

Q9: Compare and contrast spermatogenesis and oogenesis.

Q10: Critically evaluate mechanisms of fertilization.

Q11: Discuss gastrulation in the sea urchin, amphibians, and birds.

Q12: Describe the development of the eye in vertebrates.



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MAIN CAMPUS

SZL 303: GENERAL GENETICS

Date: 23rd February, 2018

Time: 8.30 - 11.30 am

INSTRUCTIONS:

- Answer ALL questions in SECTION A and any THREE from SECTION B



SECTION A: (40 MARKS)

Answer all questions, each question carries 5 marks

- 1) State the five characteristics of extra-chromosomal inheritance
- 2) The twin concept of linkage and recombination help explain the genetic underpinnings of evolution. Explain
- 3) Distinguish between the following:
 - (a) Linkage and crossing-over
 - (b) Chromosome mapping and chromosome walking
- 4) Outline the stages of restriction fragment length polymorphism (RFLP) as applied in genetic analysis.
- 5) Explain why mitochondrial DNA evolution differs from that of nuclear DNA.
- 6) Explain any two applications of polyploidy in aquaculture.
- 7) State any two characteristic of the genetic code and explain why it is said to be universal.
- 8) Briefly explain the significance of the identification of quantitative trait locus (QTL) in polygenic inheritance.

SECTION B (60 Marks)

Answer any three questions. Each question carries 20 Marks

- 9) Discuss the various aberrations in chromosome structure and their genetic consequences in humans
- 10) Give a detailed account of any five gene interactions and exceptions to Mendelian inheritance.
- 11) Discuss the advances and approaches (techniques) to gene therapy in a named genetic disorder.
- 12) Describe the structure and gene regulation of *lac* operon in *Escherichia coli*.
- 13) Genome sequencing has revolutionized Biology. Discuss



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MAIN CAMPUS

SZL 305: MOLECULAR BIOLOGY

Date: 16th February, 2018

Time: 8.30 - 11.30 am

INSTRUCTIONS:

- Answer ALL questions in SECTION A and ANY THREE questions from SECTION B.



SZL 305: MOLECULAR BIOLOGY

Date :

Time: 2 HRS

Instructions

1. Answer **ALL** questions in section A (40 marks) and **ANY THREE** in section B (60 marks)
2. Illustrate your answer(s) with labelled diagrams where appropriate

SECTION A (40marks). Each question carry equal marks (5marks)

- Q1. Briefly explain the five properties of cloning vectors
- Q2. Define the following terms or concepts: (a) translation, (b) transcription (c) genetic code, (d) codon and (e) anticodon
- Q3. Define mutation and list the possible forms of mutations
- Q4. Write short notes on gel electrophoresis and its application in Molecular Biology
- Q5. Briefly explain any five factors that support the double stranded nature of a DNA molecule
- Q6. List any four possible uses of recombinant DNA technology and biotechnology
- Q7. Briefly describe the genetic basis of frame shift and wobble effect
- Q8. Citing specific examples, briefly highlight the types of restriction enzymes and their functions

SECTION B (60 marks) Answer ANY THREE questions in this section

Each question carry equal marks (20 marks)

- Q9. Discuss the process of protein synthesis
- Q10. Discuss the significance of reverse transcriptase enzyme in complementary DNA and retroviruses
- Q11. Discuss the use of molecular biology in population genetic studies
- Q12. Critically analyse the concept of reading frames in molecular biology
- Q13. Discuss the process of achieving discontinuous DNA strand in Molecular Biology



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MAIN CAMPUS

SZL 307: COMPARATIVE ENDOCRINOLOGY

Date: 15th February, 2018

Time: 8.30 - 11.30 am

INSTRUCTIONS:

- Answer ALL questions in SECTION A and ANY TWO questions from SECTION B.



SZL 307: COMPARATIVE ENDOCRINOLOGY

TIME: 2 Hours

Section A (40 Marks): Answer all questions. Each question carries 5 marks.

- Q1: Outline the endocrine hierarchy of a given reproductive and metabolic hormone in vertebrates.
- Q2: Justify the sentence "the female sex is the default sex".
- Q3: Summarize key differences between the genomic and non-genomic modes of hormone action.
- Q4: Briefly describe the etiological differences between Type I and Type II diabetes.
- Q5: Briefly describe FIVE characteristics of feedback systems.
- Q6: Arnold Berthold is credited as one of the founders of the discipline of endocrinology. Briefly describe the experiment that earned him this credit.
- Q7: List THREE hormones secreted by the adrenal cortex and TWO hormones secreted by the adrenal medulla.
- Q8: Outline FIVE physiological events associated with the fight or flight response.

Section B (30 Marks): Answer any TWO questions. Each question carries 15 marks

Q9: Using a specific hormone example in each case, describe negative and positive endocrine feedback.

Q10: Critically evaluate the endocrinology of the stress response in vertebrates.

Q11: Compare and contrast hormonal regulation of metabolism in a named invertebrate and a named vertebrate.

Q13: Answer the following questions

- a) Briefly describe the hypothalamo-hypophyseal portal system.
- b) Discuss the development of internal and external genitalia in mammals.



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MAIN CAMPUS

SZL 309: INSECT SCIENCE

Date: 22nd February, 2018

Time: 8.30 - 11.30 am

INSTRUCTIONS:

- Answer ALL questions in SECTION A and any THREE from SECTION B
- Use illustrations where appropriate



SECTION A (40 MARKS)

ANSWER ALL QUESTIONS FROM THIS SECTION (5 MARKS EACH)

1. With **five** points distinguish between orders Lepidoptera and Odonata morphologically and reproductively and give one example of each.
2. Explain how insects exhibit **four** modes of communication giving example in each case
3. Briefly describe **three** important purposes served by excretion in the class Insecta.
4. State any **five** insect mouthpart types exhibited among insect and explain how they function and what they feed on.
5. Explain how any **three** named organs are used in respiration among the insects using specific examples
6. Briefly explain how any **three** internal and external factors control hibernation among insects
7. Explain how any **three** environmental factors that influence reproduction of the insects
8. Explain four ways by which the insects perceive their environment giving the respective organs used.

SECTION B (60 MARKS)

ANSWER THREE QUESTIONS FROM THIS SECTION. ALL QUESTIONS CARRY EQUAL MARKS (20 MARKS EACH)

9. Discuss the role of insects in (a) ecological services (b) economics of humans
10. Describe with examples the role of insect exoskeleton in the preservation and subsequent success of the insects.
11. “Insects are the only invertebrates with wings”. Discuss how this has helped the insects to survive up to today
12. “Diverse reproductive strategies have made the Class Insecta masters of their ecosystems”. Discuss this statement.
13. Discuss (i) advantages and (ii) disadvantages of being each of the following:-
 - (a) Holometabolous insect; and
 - (b) Hemimetabolous insect



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MAIN CAMPUS

SZL 311: NATURE SURVEY I

Date: 12th February, 2018

Time: 3.30 - 6.30 pm

INSTRUCTIONS:

- Answer ALL questions in SECTION A and any TWO from SECTION B.



SZL 311: NATURE SURVEY 1

Date:

Time: 3 Hrs

INSTRUCTIONS:

1. Answer ALL questions in section A (4 marks each).
2. Answer any TWO questions in Section B (15marks each).
3. Use illustrations where possible.

SECTION A: ANSWER ALL QUESTIONS IN THIS SECTION (4 MARKS EACH)

1. Briefly explain the terms: 'bird surveys'.
2. State any four adverse impacts of 'mangrove ecosystem harvesting'.
3. Explain any four factors that favour the presence of monkeys in the Maseno University environment.
4. Briefly describe eutrophication in the aquatic environments.
5. Briefly discuss the importance of any four physico chemical parameters to a named organism.
6. Briefly discuss the key environmental issues to a lake in a desert expanse.
7. Define the term overfishing, briefly stating its possible impacts.
8. Describe briefly any four ways to sample terrestrial insects live.
9. Define the term: ecotype, giving three examples.
10. Briefly describe any adaptations for the terrestrial organisms.

SECTION B: ANSWER ANY TWO QUESTIONS (15 MARKS EACH).

11. Discuss the impacts of domestic sewage pollution on the world's oceans.
12. Give a detailed account of the biota of any three named biomes.
13. Discuss the possible objectives and requirements for fish surveys.
14. Discuss the impact of human activities on any named terrestrial ecosystem.



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MAIN CAMPUS

SZL 313: APPLIED ECOLOGY

Date: 15th February, 2018

Time: 12.00 - 3.00 pm

INSTRUCTIONS:

- Answer ALL questions in SECTION A and any THREE questions in SECTION B.

